PROJECTIVE GLOBAL SECTIONS AFTER AN ELEMENTARY ETALE NEIGHBOURHOOD FOR A FINITELY PRESENTED MODULE FLAT AT A POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

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Allowed standard prerequisites (editorial).

- Openness of flatness. More on Morphisms, label more-morphisms-theorem-openness-flatness, invoked at proof line 3312. Lets the module be assumed flat over the whole of an open neighbourhood of the point, which is the first reduction.
- Existence of a complete devissage after an elementary etale neighbourhood. Same chapter flat.tex, label proposition-existence-complete-at-x, invoked at proof line 3315. Produces the diagram of pointed schemes whose horizontal arrows are elementary etale neighbourhoods and over which the pulled back module has a complete devissage at the lifted point.
- Permanence of flatness, the up and down lemma for an etale cover, the identification of the affine description, and the consequences of a complete devissage. Morphisms, label morphisms-lemma-flat-permanence (proof line 3319); Same chapter flat.tex, labels lemma-etale-flat-up-down (proof line 3321, claimed as S14-algebra-etale-flat-up-down), remark-same-notion (proof line 3323) and lemma-complete-devissage-flat-finitely-presented-module (proof line 3332, claimed as S14-algebra-devissage-fp-flat). They keep flatness after the etale base change, transfer it to the new base, translate the devissage into the affine-algebra statement, and then conclude projectivity of the module over the new base after replacing it by a standard open neighbourhood.

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Source reference. Original lemma 05IB: sources/stacks-flat/pinned-original-source.tex, source lines 3229–3229. Navigation pointer for \ref{lemma-complete-devissage-flat-finitely-presented-module} used inside the counted span (line 3332); the argument that consumes it is retained in the span.

Source reference. Original lemma 05B9: sources/stacks-flat/pinned-original-source.tex, source lines 124–124. Navigation pointer for \ref{lemma-etale-flat-up-down} used inside the counted span (line 3321); the argument that consumes it is retained in the span.

Source reference. Original theorem 0399: sources/stacks-chow/tags.txt, source lines 4264–4264. Navigation pointer for \ref{more-morphisms-theorem-openness-flatness} used inside the counted span (line 3312); the argument that consumes it is retained in the span.

Source reference. Original lemma 02JZ: sources/stacks-chow/tags.txt, source lines 3354–3354. Navigation pointer for \ref{morphisms-lemma-flat-permanence} used inside the counted span (line 3319); the argument that consumes it is retained in the span.

Source reference. Original proposition 05HR: sources/stacks-flat/pinned-original-source.tex, source lines 1429–1429. Navigation pointer for \ref{proposition-existence-complete-at-x} used inside the counted span (line 3315); the argument that consumes it is retained in the span.

Source reference. Original remark 05I0: sources/stacks-flat/pinned-original-source.tex, source lines 1731–1731. Navigation pointer for \ref{remark-same-notion} used inside the counted span (line 3323); the argument that consumes it is retained in the span.

Proposition 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $x \in X$ with image $s \in S$. Assume that*

(1) *f is locally of finite presentation,*
(2) *$\mathcal{F}$ is of finite presentation, and*
(3) *$\mathcal{F}$ is flat at x over S.*

Then there exists a commutative diagram of pointed schemes

$$\begin{array}{ccc} (X, x) & \xleftarrow{g} & (X', x') \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (S', s') \end{array}$$

whose horizontal arrows are elementary étale neighbourhoods such that X', S' are affine and such that $\Gamma(X', g^*\mathcal{F})$ is a projective $\Gamma(S', \mathcal{O}_{S'})$-module.

Proof. By openness of flatness, see More on Morphisms, Theorem 0399 we may replace X by an open neighbourhood of x and assume that $\mathcal{F}$ is flat over S. Next, we apply Proposition 05HR to find a diagram as in the statement of the proposition such that $g^*\mathcal{F}/X'/S'$ has a complete dévissage over s'. (In particular S' and X' are affine.) By Morphisms, Lemma 02JZ we see that $g^*\mathcal{F}$ is flat over S and by Lemma 05B9 we see that it is flat over S'. Via Remark 05I0 we deduce that

$$\Gamma(X', g^*\mathcal{F})/\Gamma(X', \mathcal{O}_{X'})/\Gamma(S', \mathcal{O}_{S'})$$

has a complete dévissage over the prime of $\Gamma(S', \mathcal{O}_{S'})$ corresponding to s'. Thus Lemma 05IB implies that the result of the proposition holds after replacing S' by a standard open neighbourhood of s'. □